Admin (Snapshot 1) [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

```c
#include<stdio.h>
int main()
{
        int n,r,i,j,k;
        printf("Enter number of processes: ");
        fflush(stdout);
        scanf("%d", &n);

        printf("Enter number of resource types: ");
        fflush(stdout);
        scanf("%d", &r);
        int alloc[n][r],max[n][r],need[n][r],avail[r];
        int finish[n],safe_sequence[n];
        int count=0,safeFound;
        printf("Enter allocation matrix:\n");
        for(i=0;i<n;i++)
        {
                for(j=0;j<r;j++)
                {
                        scanf("%d",&alloc[i][j]);
                }
        }
        printf("Enter max matrix:\n");
        for(i=0;i<r;i++)
        {
                for(j=0;j<r;j++)
                {
                        scanf("%d",&max[i][j]);
                }
        }
        printf("Enter available resources:\n");
        for(i=0;i<r;i++)
        {
                scanf("%d",&avail[i]);
        }
        for(i=0;i<n;i++)
        {
                finish[i]=0;
        }
        for(i=0;i<n;i++)
        {
                for(j=0;j<r;j++)
                {
                        need[i][j]=max[i][j]-alloc[i][j];
```

-- INSERT -- 1,18 Top

Mouse integration ...

Right

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Activities

Mar 21 19:11

Type to search

```
            finish[i]=0;
        }
        for(i=0;i<n;i++)
        {
            for(j=0;j<r;j++)
            {
                need[i][j]=max[i][j]-alloc[i][j];

            }
        }
        while(count<n)
        {
            safeFound=0;

            for(i=0;i<n;i++){


                if(finish[i]==0)
                {
                    int safe=1;
                    for(j=0;j<r;j++)
                    {
                        if(need[i][j]>avail[j])
                        {
                            safe=0;
                            break;
                        }
                    }
                    if(safe)
                        {
                        for(k=0;k<r;k++)
                        {
                            avail[k]=avail[k]+alloc[i][k];
                        }

                    safe_sequence[count]=i;
                    finish[i]=1;
                    safeFound=1;
                    count++;

                    }
                }
            }
-- INSERT --                                                    77,3-17       69%
```

Mouse integration ...

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File Machine View Input Devices Help

Mouse integration ...

```
            safeFound=0;

            for(i=0;i<n;i++){


                if(finish[i]==0)
                {
                        int safe=1;
                        for(j=0;j<r;j++)
                        {
                                if(need[i][j]>avail[j])
                                {
                                        safe=0;
                                        break;
                                }
                        }
                        if(safe)
                                {
                                for(k=0;k<r;k++)
                                {
                                        avail[k]=avail[k]+alloc[i][k];
                                }

                        safe_sequence[count]=i;
                        finish[i]=1;
                        safeFound=1;
                        count++;

                        }
                }
            }
            if(!safeFound)
            {
                printf("system is in unsafe state! may occur the deadlock\n");
                return 0;
            }
        }
        printf("safe sequence is:");
        for(i=0;i<n;i++)
        {
            printf("p%d",safe_sequence[i]);
        }
        return 0;
}
-- INSERT --                                                    56,2-9        94%
```

Admin (Snapshot 1) [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Activities Terminal Mar 21 16:59

sadhana@localhost:~

```
[sadhana@localhost ~]$ vi banker.c
[sadhana@localhost ~]$ gcc banker.c
[sadhana@localhost ~]$ ./a.out
Enter number of processes: 3
Enter number of resource types: 3
Enter allocation matrix:
1 1 0
0 0 1
0 0 1
Enter max matrix:
7 5 3
5 5 3
3 2 2
Enter available resources:
1 1 0
1 1 2
Safe sequence is: p0p1

[sadhana@localhost ~]$
```

Mouse integration ...

Auto capture keyboard ...